

Supplementary Information

Tailoring of defect luminescence in CVD grown monolayer MoS₂ film

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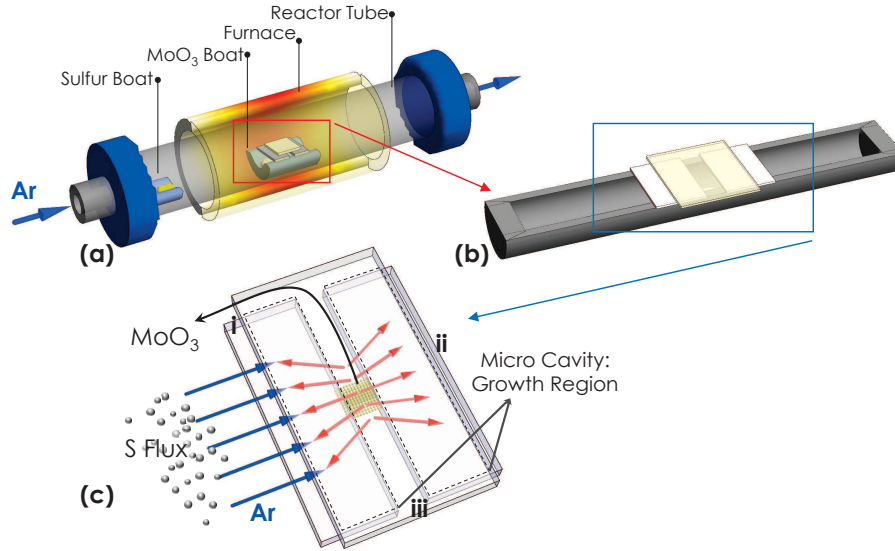


Fig. S1: Schematic representation of (a) the CVD set up used to grow MoS₂ on various substrates. It shows the placement of sulfur and MoO₃ boats inside the quartz reactor tube. MoO₃ boat was placed in the hottest part of the furnace. Sulfur boat was placed in a way, that it remains at $\sim 200^\circ\text{C}$ during growth. (In this depiction a semi-cylindrical portion of the tubular furnace has been made transparent to give a better perspective.) (b) Placement of the substrates above the MoO₃ boat. (c) An elaborate depiction of the unique substrate arrangement along with the flow pattern of different precursors. Blue arrows with the gray particles represents the Sulfur flux carried by Ar gas while the red arrows indicates the MoO₃ flux. Substrate (i) and (ii) were placed just on top of the boat with their polished surface facing up. And substrate (iii) was positioned above (i) and (ii) with its polished surface facing downward. Micro-cavity formed between (i)-(iii) and (ii)-(iii) acts as a natural reactor cavity, resulting in the formation of continuous MoS₂ monolayer, occupying the entire covered-region of all three substrates (i), (ii) and (iii) (enclosed by the dotted lines) [1]. A few of the grown layers were annealed in sulfur environment at two different temperatures 200°C and 400°C for 120 min. under the flow of argon at 4 sccm. Annealing was carried out in a similar but a separate reactor chamber. Samples were kept on top of a cleaned alumina boat with the polished surface facing downward during annealing. Sulfur was kept in a separate boat, where the temperature was maintained at $\sim 200^\circ\text{C}$.

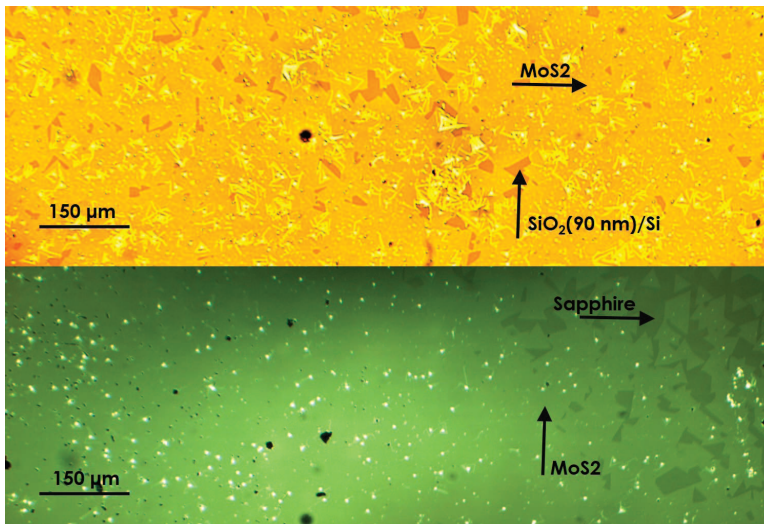


Fig. S2: Optical image of monolayer MoS_2 film grown on SiO_2/Si (top) and sapphire substrate (bottom). In both cases, growth of continuous films is evident from the color contrast. Clearly, occurrence of grain boundaries/wrinkles/foldings are much higher in the film grown on SiO_2/Si .

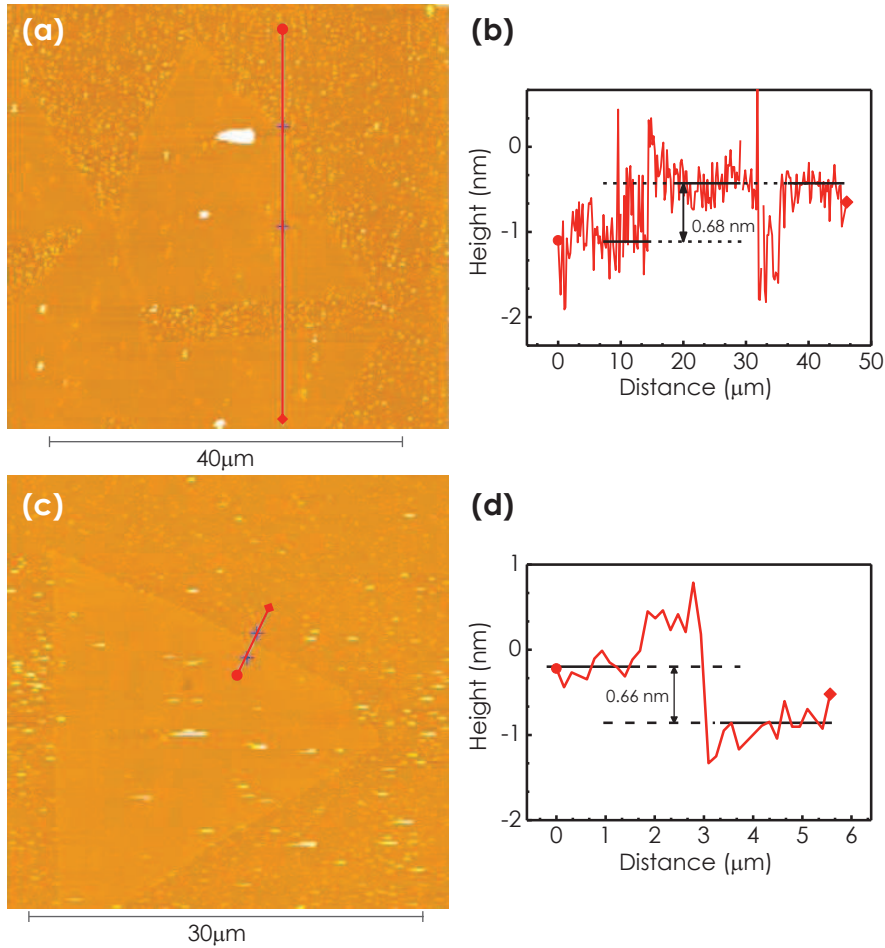


Fig. S3: Film thickness of the films grown on a sapphire substrate has been studied using atomic force microscopy (AFM) in tapping mode. (a) AFM image from an area of partial coverage, close to the edge of continuous film. (b) Height profile along the red line in figure (a). An average step height of 0.68 nm clearly suggest the synthesis of monolayer MoS₂. The same has been performed (c) on an isolated triangular flake and the line scan profile has been shown in (d).

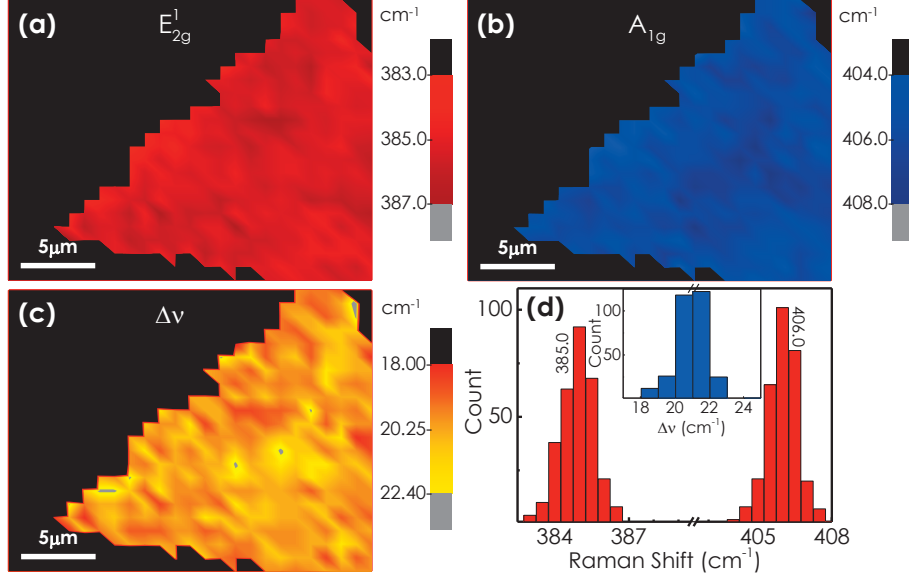


Fig. S4: Raman mapping, performed on a isolated triangular monolayer flake (similar to a flake shown in figure S3(c)). (a) and (b) show the mapping of the entire region with the positions of E_{2g}^1 and A_{1g} peaks, respectively. (c) $\Delta\nu$ map of the same region. (d) Distribution of E_{2g}^1 and A_{1g} peak positions over the mapped region. Maximum of the distributions occur, respectively, at 385.0 and 406.0 cm^{-1} . Inset shows the distribution of $\Delta\nu$ for the same area. Analysis of these histograms shows that the average and standard deviation of the distribution for the E_{2g}^1 peak are 384.9 and 0.6 cm^{-1} , respectively, while these numbers are 406.0 and 0.5 cm^{-1} , respectively, for the A_{1g} peak. It should be noted that unlike our previous work (figure 1 in the main article) the average E_{2g}^1 peak positions in this case are closer to that is typically obtained (384 cm^{-1} [2, 3]) for exfoliated monolayer samples. Moreover, the distribution of E_{2g}^1 peak is slightly narrower in the present case which can be attributed to less number of sampling (since the area is much smaller). But, interestingly A_{1g} peak distribution goes through a more pronounced change in terms of standard deviation. It can be attributed to the pristine quality of the triangular flake. Notably, number of gray regions which are earlier addressed as a overgrown particle on the surface (Fig. 1(d) in main article), is negligible in case of this triangle. Thus the perturbation on the out of plane vibrational mode is much less and so as the broadening.

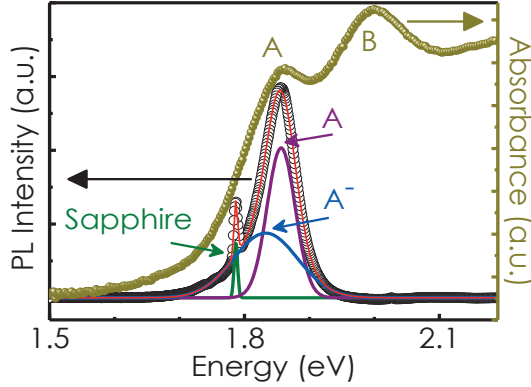


Fig. S5: Room temperature absorption and PL spectra along with three deconvoluted Gaussian peaks of the monolayer MoS_2 samples grown on sapphire. The PL spectrum is featured by a sharp peak appearing at 1.78 eV and a broader feature located at 1.85 eV. While, 1.78 eV feature has been found to be stemming from the sapphire substrate, the higher energy peak can be assigned to monolayer MoS_2 . A careful observation reveals that 1.85 eV peak is in fact consisting of two features. A fitting of the whole spectrum with three Gaussians (two for the broad feature and one for the sapphire peak) yields the positions of the constituent peaks as shown in figure S5. The two peaks, which constitute the broad 1.85 eV feature, can be attributed to free A-exciton that appears at 1.86 eV and A-trion (A^-) appearing at 1.84 eV [1, 4]. Figure S5 also shows the absorption spectrum for the same sample recorded at room temperature. Spin-split valance band edge at K symmetry points gives rise to a higher energy peak at ~ 2.00 eV (namely the B-exciton) in addition to the A-exciton peak (at 1.86 eV) [5]. The scatter plots show the experimental data. The continuous red curve represents the overall fitting of the luminescence profile.

References

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